PHYS114 Reference

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1 Data Files, Software, and Figures

1.1 Data Files

- **Binary**:
  - Advantages: compact file size.
  - Disadvantages: need encoders (otherwise difficult to read).
  - Microsoft 365 files (.xlsx, .docx, etc.), which are proprietary but have been open-sourced for interoperability with, for example, Apple and Google's suites.
  - **.fits**: Well-known, well-defined, and used by the astronomy community.
  - **.cdf**: Common Data Format, standardized encoding.
  - **.sav**: IDL's proprietary format.
- **ASCII**:
  - Advantages: built-in encoding table and, thus, easy to read.
  - Disadvantages: large file size.
  - .txt, .csv, etc.

1.2 Software

- A wide array of scientific software exists such as MATLAB and IDL.
- However, these tend to require expensive licenses.
- In contrast, Python is free and open-source.
- However, Python comes with no warranty, meaning that if something goes wrong, the user is responsible.

1.3 Figures

A professional-quality figure

- is grayscale, relying on various marker sizes and shapes (e.g. circle, diamond, etc.) as well as line styles (solid, dashed, etc.) to differentiate different series;
- contains a title, axes labels with units, and a legend as necessary;
- uses space on the plot effectively;
- has proper axis scaling; and
- is high resolution (usually 600 DPI or more).

2 Introduction to Research

Some information is specific to the New Jersey Institute of Technology (NJIT).

- A typical external federal grant is $130,000 per year for 3 years.
- Professors must bring in grants to be tenured.
- Professors also have a "2 + 2" teaching requirement: 2 courses in the fall, and 2 courses in the spring.
- If a professor supports a research assistant (RA), covering their **stipend**, **tuition**, and **fringe**, they go down to 1 + 2.
- Supporting another RA reduces this to 1 + 1.
- In contrast, teaching assistants (TAs) are paid for by the university, not the professor.
- A researcher or RA typically cannot have another job if they are already being funded by a grant for maximal dedication.
- Usually, grant money goes to the university, not the professor directly.
- PhDs in the United States usually take 6 years because new knowledge must be created to attain it.
- A typical physics graduate student coursework:
  - Fall: Electricity and Magnetism I, Classical Mechanics, Mathematical Methods
  - Spring: Electricity and Magnetism II, Quantum Mechanics, Statistical Mechanics / Thermodynamics

- Qualifying exams in the summer (June for NJIT) determine whether a student remains in the graduate program.
- Qualifying exams are difficult (typically 50% pass rate for NJIT).
- There is one last redemption exam in January for those who fail.
- In the 1980s, program officers would evaluate grant proposals, but from 1990s onward, panels of qualified individuals perform this task.

2.1 Federal Funding Agencies

- Top agencies that fund grants:
  - National Science Foundation (NSF)
  - Department of Energy (DOE)
  - National Aeronautics and Space Administration (NASA)
  - National Institutes of Health (NIH)
  - Department of Defense (DoD)
- All except NSF require **closure**, which means meaningful results must be found, and the research question must be answered.
- Some agencies require an American citizenship for all researchers.

2.2 Grant Format

- A small part of the grant ($\sim 10\%$) is:
  - Project summary (1 page)
  - Project description (15+ pages)
  - Export control, etc.
- But the majority of the grant ($\sim 90\%$) is paperwork

2.3 Fringe and Overhead

- **Fringe** is the cost of employing someone, which includes associated costs such as insurance, retirement, etc.
- **Overhead** is the money used to support the research enterprise of an institution, which includes the cost of utilities, supplies, subscriptions, etc.
- The fringe at NJIT went from 43% to 86% in 5 years.
- The cost of research has skyrocketed due to rising insurance premiums and unionization.

3 Forward Model and Photometer

3.1 Forward Model

A mathematical or quantitative description of the problem you're solving. One can use synthetic data and example numbers to test.

3.2 Photometer

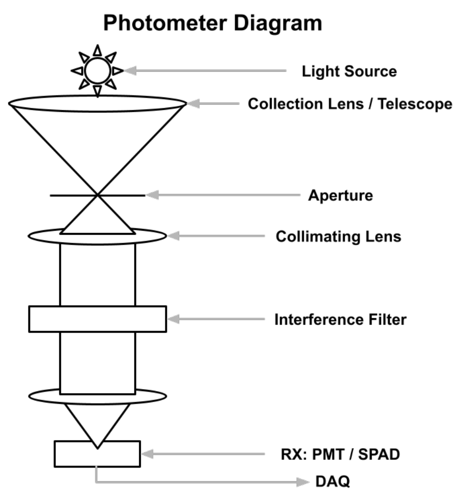


Figure 1: A diagram of a photometer.

- Measures the number of photons from a source.
- IF stands for interference filter.
- RX stands for receiver.
- TX stands for transmitter.
- The receiver can either be a **photomultiplier tube (PMT)** or a **single-photon avalanche diode (SPAD)**. A PMT is a single-element instrument that counts individual photons precisely.
- **Charge-coupled devices (CCDs)** are an array of sensors that take an image of incoming photons. Scientific-grade CCDs are more expensive than PMTs.
- **Photon count formula**:

  $$N = N_{\text{signal}} + N_{\text{noise}}$$

  where

- N is the total number of photons.
- N_{signal} is the number of photons from the desired source, and
- N_{noise} is the number of photons from other (undesirable) sources.

- **Noise photon count formula**:

$$N_{\text{noise}} = N_C \Delta t$$

where

  - N_C [counts/s] is the noise count rate, and
  - $\Delta t\,[s]$ is the **integration time** (the time interval at which the detector collects photons).

- **Signal photon count formula**:

$$N_{\text{signal}} = BA\Delta t\,\Omega T_A \nu_{RX}$$

$$\Omega = \frac{A}{r^2} = 2\pi(1 - \cos\alpha)$$

where

  - $B\,[\text{photons}/(\text{s}\cdot\text{m}^2\cdot\text{str})]$ is a constant.
  - $A\,[\text{m}^2]$ for N_{signal} is the area of the telescope.
  - $\Delta t\,[s]$ is the integration time.
  - $\Omega\,[\text{str}]$ is the solid angle.
  - T_A is the atmospheric transmission.
  - ν_{RX} is the receiver efficiency.
  - A for N_{noise} is the area of the detector.
  - r is the focal length of the telescope.
  - α is the half-angle of the cone of light collected by the telescope, also known as the field of view.

Note: $A\Omega$ is a conversed quality called the **etendue**, and it will be explored later.

4 Distributions and Uncertainty

- For **nonstationary** PDFs, the moments are in terms of time.
- For **wide-sense stationary (WSS)** PDFs, only the **first two moments** are stationary, which represents most real-life phenomena.
- For **stationary** PDFs, all the moments are constant and *not* in terms of time.

- **Moment generating function (continuous)**:

$$n^{\text{th}} \text{ moment} = \int_{-\infty}^{\infty} x^n \, \text{PDF}(x) \, \partial x$$

- **Moment generating function (discrete)**:

$$n^{\text{th}} \text{ moment} = \sum_{i=1}^{\infty} x_i^n \, \text{PDF}(x_i)$$

- This class only focuses on the Gaussian, Binomial, and Poisson distributions. The rest are extra information.
- Remember that probabilities multiply. The probability of reproducing an entire dataset is

$$\prod_{i=1}^{N} \text{PDF}(x_i) = p_1 p_2 \ldots p_N$$

- General process of data collection and processing:
  1. Determine PDF.
  2. Create estimators for moments from data.
  3. Determine if estimators are biased or unbiased using **expectation values**.
  4. Determine the chi-squared goodness of fit.
- From strongest to weakest: independent, uncorrelated, and no covariance.
  - **Independent**: no true mathematical representation of independence.
  - **Uncorrelated**: $r \equiv \frac{\sigma_{XY}^2}{\sigma_X \sigma_Y} = 0$
  - **No covariance**: $\sigma_{XY}^2 = 0$

4.1 Gaussian Distribution

Also known as the normal distribution. Represents humans and many natural phenomena.

$$f(x; \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right]$$

- μ and σ are parameters.
- 1^{st} moment (mean): μ
- 2^{nd} moment (variance): σ^2

- Estimator for 1st moment:

$$\bar{x} = \frac{1}{N}\sum_{i=1}^{N} x_i$$

- Estimator for 2nd moment:

$$s^2 = \frac{1}{N-1}\sum_{i=1}^{N}(x_i - \bar{x})^2$$

- 3rd moment (skewness)
- 4th moment (kurtosis)
- Uncertainty for 1st moment:

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{N}}$$

  This signals diminishing returns for increasing N.

- However, the above is assuming WSS. In a case without WSS:

$$\bar{x} = \frac{\sum_{i=1}^{N}\left(\frac{x_i}{\sigma_i^2}\right)}{\sum_{i=1}^{N}\left(\frac{1}{\sigma_i^2}\right)}$$

$$\sigma_{\bar{x}}^2 = \frac{1}{\sum_{i=1}^{N}\frac{1}{\sigma_i^2}}$$

4.2 Binomial Distribution

$$\mathrm{PDF}(x; n, p) = \binom{n}{x} p^x (1-p)^{n-x}$$

where

$$\binom{n}{x} = \frac{n!}{x!(n-x)!}$$

- Trials must be **independent**.
- For p large and n large, similar to the **Gaussian distribution**.
- For p low and n large, similar to the **Poisson distribution**.
- n and p are parameters.
- 1st moment: np

- 2^{nd} moment: $np(1-p)$
- 3^{rd} moment: $\frac{(1-p)-p}{\sqrt{np(1-p)}}$
- Estimator for 1^{st} moment:

$$\bar{x} = \frac{1}{N}\sum_{i=1}^{N} x_i$$

- Estimator for 2^{nd} moment:

$$s^2 = \frac{1}{N-1}\sum_{i=1}^{N}(x_i - \bar{x})^2$$

4.3 Poisson Distribution

Represents scenarios with counting and waiting for events.

$$\text{PDF}(x;\lambda) = \frac{\lambda^x e^{-\lambda}}{x!}$$

- λ is the parameter.
- 1^{st} moment: λ
- 2^{nd} moment: λ
- 3^{rd} moment: $\frac{1}{\sqrt{\lambda}}$
- Estimator for 1^{st} moment:

$$\bar{x} = \sum_{i=1}^{N} x_i$$

- Estimator for 2^{nd} moment:

$$s^2 = \sum_{i=1}^{N} x_i = \bar{x}$$

- Percent error:

$$\frac{1}{\sqrt{\bar{x}}} \times 100\%$$

- SNR:

$$\frac{\bar{x}}{\sigma} = \sqrt{\bar{x}}$$

4.4 Laplace Distribution

$$f(x;\mu,b) = \frac{1}{2b}\exp\left[-\frac{x-\mu}{b}\right]$$

- μ and b are parameters.
- μ is a location parameter.
- $b > 0$ is called "diversity."
- 1^{st} moment: μ
- 2^{nd} moment: $2b^2$
- 3^{rd} moment: $\emptyset$

4.5 Gamma Distribution

$$f(x;\alpha,\beta) = \frac{\beta^\alpha}{\Gamma(\alpha)}x^{\alpha-1}e^{-\beta x}$$

where Γ is the gamma function.

- α and β are parameters.
- α is the shape parameter.
- $\frac{1}{\beta}$ is the scale parameter.
- 1^{st} moment: $\frac{\alpha}{\beta}$
- 2^{nd} moment: $\frac{\alpha}{\beta^2}$
- 3^{rd} moment: $\frac{2}{\sqrt{\alpha}}$

4.6 Assumptions

In this class, we assume that

1. Stationary or WSS PDF: $\sigma_i \approx \sigma$
2. Each measurement is independent and identically distributed (IID).
3. **Method of Maximum Liklihood (MML)**
   - Also known as Maximum Likelihood Method (MLM).
   - Maximum probability of reproducing the data.
   - An alternate method is **Maximum Entropy Method (MEM)** that seeks to add as much disorder as possible to the datapoints, but this class only focuses on MML.
4. An implicit assumption is the zero-mean process. We assume or subtract off the mean to make calculations much easier.

4.7 Expectation Values

$$\langle x_i \rangle = E[x_i] = \mu \quad \text{(typically)}$$

Examples:

- We expect the $\bar{x}$ estimator to yield μ: $\langle \bar{x} \rangle = \mu$
- And similarly for s^2: $\langle s^2 \rangle = \sigma^2$

A rule for expectation values is that you ignore constants:

$$\langle c \rangle = c$$

You can also ignore summations:

$$\langle \sum_{i=1}^{N} x_i \rangle = \sum_{i=1}^{N} \langle x_i \rangle$$

4.8 Uncertainty Propagation

4.8.1 Basics

Uncertainty is represented by σ. For example, the uncertainty in $\bar{x}$ is denoted as $\sigma_{\bar{x}}$. Uncertainty can either propagate in a **worst case** or **even-steven** (informally named by our professor) scenario.

4.8.2 Worst Case Propagation

Basics for worst case propagation: Suppose there are height measurements h_1 and h_2 with uncertainties σ_{h_1} and σ_{h_2}. Adding the two heights together also adds their uncertainties: $(h_1 \pm \sigma_{h_1}) + (h_2 \pm \sigma_{h_2}) = (h_1 + h_2) \pm (\sigma_{h_1} + \sigma_{h_2})$. This is the simplest example, but this can get much more complex, especially for operations like multiplication.

4.8.3 "Even-Steven" Propagation

The Taylor series approximation of the "even-steven" propagation of uncertainty:

$$\sigma_x^2 \approx \sigma_u^2 \left(\frac{\partial x}{\partial u}\right)^2 + \sigma_v^2 \left(\frac{\partial x}{\partial v}\right)^2 + \cdots + 2\sigma_{uv}^2 \left(\frac{\partial x}{\partial u}\right)\left(\frac{\partial x}{\partial v}\right) + \ldots$$

$$\sigma_{uv}^2 = \frac{1}{N-1} \sum_{i=1}^{N} (u_i - \bar{u})(v_i - \bar{v})$$

where σ_{uv}^2 is the covariance, which is 0 when both u and v are **independent**.

4.8.4 "Even-Steven" Application

The "even-steven" propagation of uncertainty formula above can be applied to derive the uncertainty $\sigma_{\bar{x}}$ of the estimator $\bar{x}$:

$$
\begin{aligned}
\bar{x} &= \frac{1}{N}\sum_{i=1}^{N} x_i \\
\sigma_{\bar{x}}^2 &= \sum_{i=1}^{N} \sigma_{x_i}^2 \left(\frac{\partial \bar{x}}{\partial x_i}\right)^2 \qquad (2) \\
\sigma_{\bar{x}}^2 &= \sum_{i=1}^{N} \sigma_{x_i}^2 \left(\frac{1}{N}\right)^2 \\
\sigma_{\bar{x}}^2 &= N\sigma^2 \left(\frac{1}{N}\right)^2 \qquad (4) \\
\sigma_{\bar{x}}^2 &= \frac{\sigma^2}{N} \\
\sigma_{\bar{x}} &= \frac{\sigma}{\sqrt{N}}
\end{aligned}
$$

Often times, the dataset contains independent measurements, leading to the covariance terms being zero in (2). Furthermore, the dataset is often stationary or wide-sense stationary, meaning that $\sigma_{x_i} \approx \sigma$ for all i, as shown in (4).

4.9 Chi-Squared

The following only applies to data that is independent and identically distributed.

4.9.1 Chi-Squared Goodness of Fit

Given a histogram:

$$\chi^2 \equiv \sum_{j=1}^{n} \left(\frac{h(x_j) - NP(x_j)}{\sigma_j(h)}\right)^2 = \sum_{i=1}^{N} \frac{(x_i - \bar{x})^2}{\sigma_i^2}$$

where

- n is the number of bins.
- $h(x_j)$ is the value of the j-th bin.
- N is the total number of measurements.
- $P(x_j)$ is the probability of the j-th bin.
- $\sigma_j(h)^2$ is the variance in the j-th bin.

χ^2 is a measure of the goodness of fit of a model to the data. A smaller χ^2 indicates a better fit. Ideal value is 0, but in practice it is around n because the numerator and denominator of the summation are around the same order of magnitude.

4.9.2 Reduced Chi-Squared

$$\chi_\nu^2 = \frac{\chi^2}{\nu}$$
$$\nu = n - n_c$$

- n_c is the number of constraints. Usually 1.
- ν is the number of degrees of freedom. Usually $n-1$ with no outliers.

4.9.3 Generalized Chi-Squared

$$\chi^2 \equiv \sum_{i=1}^{N} \frac{(y_i - y(x_i))^2}{\sigma_i^2}$$

where

- y_i is the value of the i-th measurement,
- $y(x_i)$ is the value of the model at the i-th measurement, and
- σ_i^2 is the uncertainty in the i-th measurement.

4.9.4 Chi-Squared Distribution

$$P_{\chi^2 > \chi_T^2}(\chi^2, \nu) = \int_{\chi^2}^{\infty} P_{\chi^2}(\chi^2; \nu)\, d\chi^2$$
$$P_{\chi^2}(\chi^2, \nu) = \frac{1}{2^{\nu/2}\Gamma(\nu/2)} (\chi^2)^{\nu/2-1} e^{-\chi^2/2}$$

where

- ν is the number of degrees of freedom,
- Γ is the gamma function, and
- χ^2 is the chi-squared value. Note that χ^2 is a single value, so do not treat it as *chi* raised to 2.

The purpose of the chi-squared distrubtion is to determine the probability that the parent model would yield a χ^2 value greater than the one calculated from the data.

On exams, a table will be given to find the probability using the chi-squared distribution.

4.10 Probability of Exceeding a Correlation Coefficient

$$P(r;N) = 2\int_{|r|}^{1} \frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\pi}\Gamma(\frac{\nu}{2})}(1-r^2)^{\frac{\nu-2}{2}}dr$$

where

- Γ is the gamma function,
- r is the correlation coefficient, and
- ν is the number of degrees of freedom. In this case, $\nu = N - 2$.

This is used to determine the probability that the sample appears correlated when it is not.

5 Remote Sensing

Remote sensing is the science of obtaining information about objects or areas from a distance, typically from aircraft or satellites. **Passive** remote sensing "watches" the system, collecting without impacting the system (e.g., photometer). **Active** remote sensing perturbs the system until it reaches steady-state again (e.g., radar). This "nudge" must be small to minimize disruption to the system. Passive remote sensing equipment is preferred as it tends to be less complex than active remote sensing equipment.

5.1 LiDAR

LiDAR, which stands for Light Detection and Ranging, is an active remote sensing method that uses pulses of infrared (or sometimes visible or ultraviolet) light to measure the system. LiDAR offers a higher resolution than radar due to using shorter wavelength of light than radio waves.

5.2 Radar

Radar, which stands for Radio Detection and Ranging, is an active remote sensing method that uses radio waves to measure the system. Radar offers a lower resolution than LiDAR due to using radio waves, which have a longer wavelength.

5.3 Range Equation

$$r = \frac{c\Delta t}{2}$$

where

- r is the range to the target,
- c is the speed of light,
- Δt is the time it takes for the light to return to the receiver.

5.4 Angle of Spread

$$\theta \sim \frac{\lambda}{D}$$

$$\theta = 1.22\frac{\lambda}{D} \qquad \text{for circular apertures}$$

where

- θ is the angle of spread,
- λ is the wavelength of the light,
- D is the diameter of the aperture.

Thus it is clear why radar has a lower resolution than LiDAR, because radio waves have a longer wavelength and thus spread out much more. This leads to a **volume underfilled** scenario, in which the target does not fill the entire volume of the beam.

5.5 Doppler Effect

$$\lambda' = \frac{v - v_s}{f}$$

$$f' = f_0'(\frac{v \pm v_o}{v \mp v_s})$$

where

- λ' is the observed shifted wavelength,
- f' is the observed shifted frequency,
- v is the speed of the wave,
- v_o is the speed of the observer,
- v_s is the speed of the source, and
- f_0' is the emitted frequency.

Sign conventions: In the numerator, use + if the observer is moving towards the source, and - if the observer is moving away from the source. In the denominator, use - if the source is moving towards the observer, and + if the source is moving away from the observer.

5.6 Monostatic and Bistatic Systems

In a **monostatic** system, the transmitter and receiver are at the same location. In a **bistatic** system, the transmitter and receiver are at different locations (usually common for speed detection in rural areas).

5.7 Speed Detection

Speed detection is a common application of remote sensing and can use either radar or LiDAR. Radar speed detection uses the doppler effect to measure the speed of a vehicle, whereas LiDAR speed detection uses the time it takes for the light to return to the sensor to calculate the vehicle's speed.

Radar has a low **signal-to-noise ratio** (SNR) compared to LiDAR at the receiver, allowing a car to detect when its speed is being tracked and jamming the reflected signal (thus, radio detectors have been outlawed in many places). LiDAR has a high SNR (especially when directed at license plates with a special coating), so while a car still detect when its speed is being tracked, it cannot jam the reflected signal (as a result, LiDAR detectors have not been outlawed).

5.8 Stealth

Stealth aircraft conceal themselves to radar detection by using a combination of radar-absorbent materials and clever geometry to deflect radar waves away from the receiver. This is why stealth aircraft are not completely invisible to radar, but they have a very low radar cross-section.

5.9 Jenny Jump

Jenny Jump uses active remote sensing to detect the height that particles are in the atmosphere. The transmitter sends a beam of green light up into the atmosphere 30 times per second. (One pulse would have a low SNR, which is solved by stacking 30 pulses.) The receiver then detects the light that is scattered back to it from the particles in the atmosphere. The height of the particles is then calculated using the time it takes for the light to return to the receiver (the range equation applied vertically).

Among other information, this receiver can detect gravity waves (which are a type of buoyancy waves) in the atmosphere. Gravity waves are a cause of clear-air turbulence, which is a hazard for aircraft.

6 Monte Carlo Methods

This class does not focus on Monte Carlo methods, but here is a brief overview. **Monte Carlo methods** are a broad class of computational algorithms that rely on repeated random sampling to estimate possible outcomes of complex, uncertain events. Random number generation is a complex topic, yet it is important for simulations. For example, the simulation of light scattering and weather models predicting a hurricane's path use Monte Carlo methods. Weather models are **ensemble models**, which means that they run the same simulation multiple times with different initial conditions to generate a range of possible outcomes and account for uncertainty. Ensemble models only help for so long until the models start diverging greatly.

7 Fitting Simple Curves

Fitting to a Line

7.1 Straight Line

Forward model:

$$y = a_0 + b_0 x$$

Let σ_i denote the uncertainty in y_i. (The uncertainty in x_i is assumed to be 0.) x_i and y_i are the data points.

Minimize $\Delta y = y_i - y(x_i)$, the difference between the data points and the model. Assume that each y_i is pulled from a stationary Gaussian distribution, and use the maximum likelihood method.

The probability of a particular observation is

$$P_i = \frac{1}{\sigma_i\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{y_i - y(x_i)}{\sigma_i}\right)^2\right].$$

For N measurements:

$$\prod_{i=1}^{N} P_i = \prod_{i=1}^{N}\left(\frac{1}{\sigma_i\sqrt{2\pi}}\right) \exp\left[-\frac{1}{2}\sum_{i=1}^{N}\left(\frac{y_i - y(x_i)}{\sigma_i}\right)^2\right].$$

For the model $y = a_0 + b_0 x$:

$$P(a_0, b_0) = \prod_{i=1}^{N}\left(\frac{1}{\sigma_i\sqrt{2\pi}}\right) \exp\left[-\frac{1}{2}\sum_{i=1}^{N}\left(\frac{y_i - (a_0 + b_0 x_i)}{\sigma_i}\right)^2\right]$$

To maximize $P(a_0, b_0)$, we need to minimize the exponent:

$$\chi^2 = \sum_{i=1}^{N}\left(\frac{y_i - (a_0 + b_0 x_i)}{\sigma_i}\right)^2$$

To minimize χ^2, we take the partial derivatives with respect to a_0 and b_0 and set them to 0.

$$\frac{\partial \chi^2}{\partial a_0} = 0 \quad \text{and} \quad \frac{\partial \chi^2}{\partial b_0} = 0.$$

The result is:

$$\Delta = \sum \frac{1}{\sigma_i^2} \sum \frac{x_i^2}{\sigma_i^2} - \left(\sum \frac{x_i}{\sigma_i^2} \right)^2$$
$$a_0 = \frac{1}{\Delta} \left(\sum \frac{x_i^2}{\sigma_i^2} \sum \frac{y_i}{\sigma_i^2} - \sum \frac{x_i}{\sigma_i^2} \sum \frac{y_i}{\sigma_i^2} \right)$$

b_0 can be found as well, but the formula is not the focus of this course. The uncertainty on a_0 and b_0 is given by:

$$\sigma_z^2 = \sum \sigma_i^2 \left(\frac{\partial z}{\partial y_i} \right)^2$$

where z can be a_0 or b_0. Note that the number of contraints n_C is 2 since due to a_0 and b_0.

7.2 Polynomials

Forward model:

$$y(x_i) = \sum_{k=1}^{m} a_k f_k(x_i)$$

where $f_k(x_i) = 1, x_i, x_i^2, \ldots$ is a polynomial parent function.

Carry out a similar process as fitting a straight line:

$$P = \prod_{i=1}^{N} \left(\frac{1}{\sigma_i \sqrt{2\pi}} \right) \exp \left[-\frac{1}{2} \sum_{i=1}^{N} \left(\frac{y_i - \sum_{k=1}^{m} a_k f_k(x_i)}{\sigma_i} \right)^2 \right]$$

Thus:

$$\frac{\partial \chi^2}{\partial a_\ell} = 0 \qquad \text{for } \ell = 1, \ldots, m$$

8 Linearization

Linearization is the process of transforming a non-linear function into a linear function:

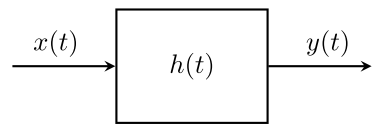
$$y = Ae^{-bx}$$
$$\ln y = \ln(Ae^{-bx})$$
$$\ln y = \ln A - bx$$

9 System Theory

System theory is the study of how individual components of a system interact with each other to produce a collective behavior. A **linear time-invariant (LTI)** system is a system that satisfies the following properties:

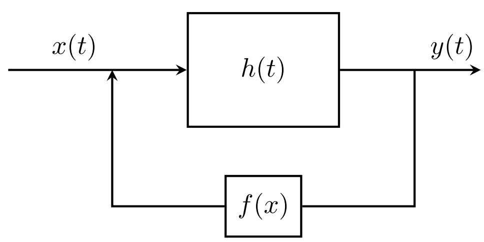
- **Linearity**: If the input is multiplied by a constant, the output is multiplied by the same constant.
- **Time-invariance**: If the input is shifted in time, the output is shifted by the same amount of time. In other words, the time at which the input is provided does not affect the system's output.

A simple diagram of a system is:



where $x(t)$ is the input, $h(t)$ is the system, and $y(t)$ is the output. The system is often considered a "black box" meaning we can only infer the properties of the system by observing the input and output.

A system can also have a feedback from the output to the input, represented by $f(x)$:



System theory involving feedback is called **control theory**.

10 Perturbation Theory

The notation from system theory carries over:

- $x(t)$ is the input
- $y(t)$ is the output
- $h(t)$ is the system

10.1 Dirac-Delta Function

$$\delta(t) \equiv \begin{cases} \infty & t = 0 \\ 0 & t \neq 0 \end{cases}$$

$$\int_{-\infty}^{\infty} \delta(t) dt = 1$$

The Dirac-Delta function is continuous, which is useful in analog applications.

10.2 Kronecker Delta Function

$$\delta_{ij} \equiv \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

The Kronecker Delta function is discrete, which is useful in digital applications such as digital signal processing (DSP).

10.3 Step Function

$$u(t) \equiv \begin{cases} 1 & t \geq 0 \\ 0 & t < 0 \end{cases}$$

$$\frac{d}{dt} u(t) = \delta(t)$$

Also known as the Heaviside step function.

10.4 Rectangle Function

$$\mathrm{rect}\left(\frac{t - t_0}{a}\right) \equiv \text{(a rectangle centered at } x = t_0 \text{ with width } a\text{)}$$

$$\mathrm{rect}\left(\frac{t - X}{Y}\right) \equiv u(t - X + Y/2) - u(t - X - Y/2)$$

10.5 Sinc Function

$$\mathrm{sinc}(x) \equiv \frac{\sin x}{x}$$

10.6 Convolutions

To get the output from the input:

$$y(t) = x(t) * h(t)$$

where $*$ is the convolution operator, defined as:

$$x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$$

Convolutions can be informally thought as ”smearing” two things together. Convolutions can be computed mathematically or graphically. In this class, we will only compute convolutions graphically.

Properties:

- When $x(t) = \delta(t)$, $y(t) = h(t)$.
- After many convolutions, the output will approach a Gaussian distribution.
- A delay circuit is when a function is shifted without distortion.

11 Fourier Transform

$$\mathcal{F}\{x(t)\} \equiv X(\omega) \equiv \int_{-\infty}^{\infty} x(t)e^{-i\omega t}dt$$

The **Fourier transform** $\mathcal{F}$ maps a signal from the time domain to the frequency (ω) domain, resulting in the signal's **spectrum**. It can also be used on spatial data. A key advantage of the Fourier transform is data compression: frequency values often take up much less storage space than a full representation of the signal in the time domain.

The convention is to use a lowercase function like $x(t)$ in the time domain and an uppercase function like $X(\omega)$ in the frequency domain.

The integral usually yields a real and an imaginary component. The imaginary component contains phase information. However, we are usually only interested in the real component since that yields frequency information.

It is important to note that there are other definitions of the Fourier transform, which differ by a constnat multiple.

11.1 Discrete Fourier Transform

$$\mathcal{F}\left\{x(n)\right\} \equiv X(k) \equiv \sum_{n=0}^{N-1} x(n) e^{-i\frac{2\pi kn}{N}}$$

where N is the number of data points.

The **discrete Fourier transform (DFT)** is suitable for computers. However, in practice, software uses the FFT.

11.2 Fast Fourier Transform

If $N = 2^q$ for some integer q, then the DFT can be computed much faster using the **fast Fourier transform (FFT)**. Whereas the regular Fourier transform and DFT take $O(N^2)$ time to compute, the FFT takes $O(N \log N)$ time to compute. Any number of data points can be brought to the next highest power of 2 by padding the ends with zeros (a technique called **zero-padding**).

11.3 Fourier Transform Pairs

There are many Fourier transform pairs, but these are the ones we will need in this class. The following table is constructed based on the definition of the Fourier transform used above. The pairs can change by a constant multiple depending on the definition used.

$x(t)$	$X(\omega)$
1	$2\pi\delta(\omega)$
$\delta(t)$	1
$\mathrm{rect}(t/T)$	$T\ \mathrm{sinc}(\frac{T}{2\pi}\omega)$
$\mathrm{sinc}(\omega t)$	$\frac{1}{\omega_0}\mathrm{rect}(\frac{\omega}{2\pi\omega_0})$
$\cos(\omega_0 t)$	$\pi[\delta(\omega-\omega_0)+\delta(\omega+\omega_0)]$
$\sin(\omega_0 t)$	$-i\pi[\delta(\omega-\omega_0)-\delta(\omega+\omega_0)]$

11.4 Convolution Theorem

$$x(t) * h(t) = \mathcal{F}^{-1}\left\{\mathcal{F}\left\{x(t)\right\}\mathcal{F}\left\{h(t)\right\}\right\}$$
$$x(t) \cdot h(t) = \mathcal{F}^{-1}\left\{\mathcal{F}\left\{x(t)\right\} * \mathcal{F}\left\{h(t)\right\}\right\}$$

Notice that multiplication in the time domain is equivalent to convolution in the frequency domain, and convolution in the time domain is equivalent to multiplication in the frequency domain. Thus, a convolution inherently changes the frequency content of a signal.

This theorem is useful in because convolutions are much slower to compute than the Fourier transform.

11.5 PSD

The PSD represents the power of each frequency.

$$\text{PSD}(\omega) = |X(\omega)|^2 = (\text{real})^2$$

PSD has multiple full-forms:

- Power Spectrum
- Power Spectrum Distribution
- Power Spectral Distribution
- Power Spectral Density

Each are technically defined differently, but are often used interchangeably. We will use PSD to avoid any confusion.

11.6 Windowing Functions

Collecting a finite number of samples in a signal (which is the experimental reality) inherently applies a rectangular window when performing the Fourier transform. This introduces **spectral leakage**, the inclusion of artificial frequencies near the ends of the data. Therefore when dealing with finite samples, it is essential to use a windowing function to mitigate the spectral leakage caused by the rectangular window.

$$\mathcal{F}\{x(t)\} = \mathcal{F}\{x(t)w(t)\} = X(\omega) * W(\omega)$$

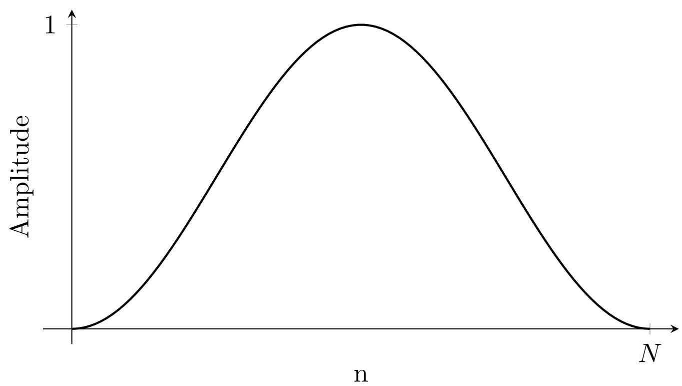
where $w(t)$ is a windowing function.

Windowing functions can be evaluated by plotting their Fourier transforms on a log scale magnitude using decibels (dB) in the frequency domain. This leads to a graph with multiple lobes. The **main lobe** is the first lobe and determines the frequency resolution, and the **side lobes** are the subsequent lobes that determine the dynamic range. The main lobe should be as narrow as possible, and the side lobes should be as low as possible, but in real life a compromise must be made. The rectangular window actually has the narrowest main lobe, but it has high side lobes. The most ideal case is with the main lobe having zero width and side lobes having zero amplitude, which is a Dirac delta spike at $\omega = 0$.

However, this is not practical in real life as it requires an infinite number of data points. Thus, it is often more common to prioritize a large main lobe to drive down the side lobes and increase the dynamic range. This allows weaker signals to be easily discerned next to strong signals. This is at the cost of frequency resolution: two closely spaced frequency spikes may be blurred together into a single spike.

We will only cover these 2 windows in this class, but many others exist (such as trapezoid, Barlett, Blackman, etc.). Nowaways, it is common to run through multiple windowing functions to see which one works best for the given application.

11.6.1 Hann Window



$$W(n) = \frac{1}{2}\left[1 - \cos\left(\frac{2\pi n}{N-1}\right)\right]$$

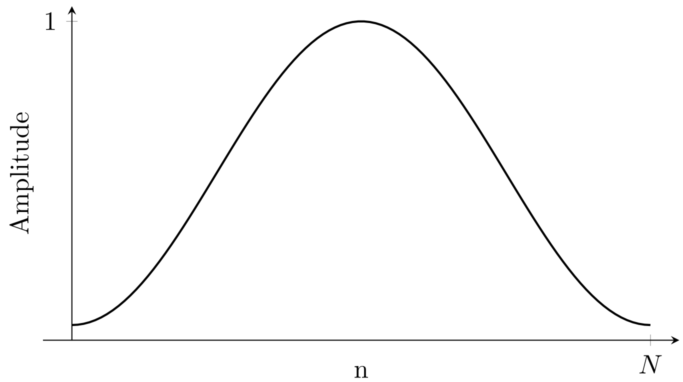
where

- N is the window length (number of data points),
- n is the index of the data point, $0, 1, \ldots, N-1$.

The Hann (or Hanning) window sacrifices frequency resolution for a lower dynamic range; that is, its main lobe is wider than the rectangular window's, but its side lobes are much lower.

A notable feature about the Hann window is that it drives the ends of the signal to 0.

11.6.2 Hamming Window



$$W(n) = 0.54 - 0.46\cos\left(\frac{2\pi n}{N-1}\right)$$

where

- N is the window length (number of data points),
- n is the index of the data point, $0, 1, \ldots, N-1$.

Unlike the Hann window, the Hamming window does not drive the ends of the signal to 0. It also has a larger main lobe and lower side lobes than the Hann window, but these side lobes remain more or less constant instead of decaying.

12 Filter Theory

12.1 Introduction

Filter theory often deals in the spectral (frequency) domain to process signals. For example, "AC hum" (often 60 Hz with a 120-Hz harmonic) is a common source of noise that can be removed by filters.

Another application of filter theory is basic image editing. For example, a picture of a tiger behind bars can be edited to remove the bars by applying a filter that removes the frequencies corresponding to the bars. Modern smartphones use such filters for photography and image processing.

12.2 Types of Filters

12.2.1 Band Stop

A band stop filter *removes* a specific range of frequencies from a signal. For "AC hum", a band stop filter would remove the frequencies around 60 Hz or 120 Hz.

12.2.2 Band Pass

A band pass filter (the opposite of a band stop filter) *passes* only a specific range of frequencies through the filter.

12.2.3 Low Pass

A low pass filter suppresses the high frequencies. In images, this applies smoothing and reduces "graininess."

12.2.4 High Pass

A high pass filter suppresses the low frequencies. In images, this sharpens the image and adds contrast.

12.3 Log-Log PSD Plots

Often times, linear plots are not helpful due to the large range of frequencies and powers, generating curves that are not easy to analyze. Log-log PSD plots use the logarithm of PSD on the y-axis and the logarithm of the frequency on the x-axis, helping visualize the PSD for a wide range of frequencies and powers via a linear curve. Noise forms a straight line on a log-log PSD plot, and signals form discernible peaks. The goal of filters is to enhance signals and suppress noise by altering the slope of the PSD plot.

The **DC offset** is the average constant value of the signal, leading to a large spike at $\omega = 0$, which can distort the plot's scale and obscure the rest of the signal. Thus, it is best to remove the DC offset before plotting the PSD in a process called **zero-meaning** the data.

12.4 Types of Noise

12.4.1 White Noise

White noise has a constant power spectral density across all frequencies, leading to a flat line on a log-log PSD plot. This is the most desirable type of noise because it is the easiest to filter out (e.g. via a band pass filter for a specific signal frequency).

12.4.2 Red Noise

Red noise has a power spectral density that dominates in the low frequencies and rapidly drops off with increasing frequency. This is tougher to filter out than white noise especially if the signal is in the low frequencies. In practice, engineers often use a process called **whitening** to convert red noise into white noise.

12.4.3 Pink Noise

Pink noise has a slope between that of white and red noise with a power spectral density that more slowly decreases with frequency. Most electronic components create pink noise.

12.5 FPGA

A **Field-Programmable Gate Array (FPGA)** is a type of integrated circuit that can be programmed to perform a wide range of functions after manufacturing it. They bridge the gap between software and hardware, allowing for high-speed, real-time signal processing.

12.6 FIR and IIR

FIR	IIR
Finite Impulse Response	**Infinite Impulse Response**
No feedback loops	Feedback loops
More stable	Less stable
Linear phase	Loses phase information
More computationally expensive	Less computationally expensive
More memory intensive	Less memory intensive
Less mathematically complex	More mathematically complex
e.g. moving average, raised cosine	e.g. Butterworth, Chebyshev

The **order** of a filter (either FIR or IIR) is the number of delay blocks used (Z^{-1}). The higher the order, the more computationally expensive the filter is.

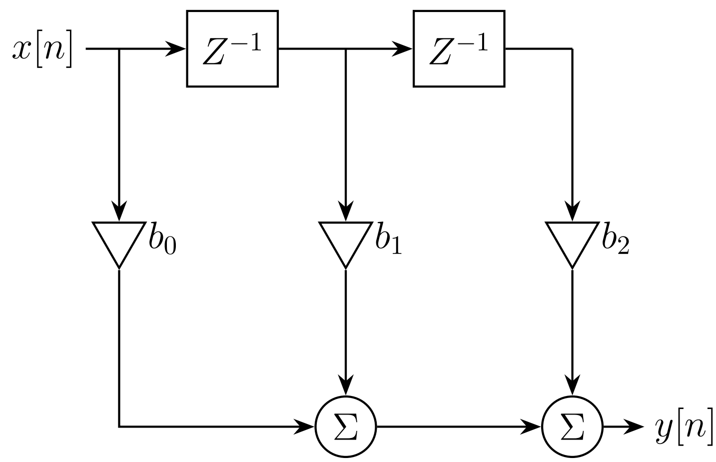
An FIR filter's response eventually goes to 0, while an IIR filter's response may propagate forever due to feedback loops. A higher-order FIR filter leads to a sharper frequency cutoff. An FIR filter is also less mathematically complex than an IIR filter, but it generally requires a higher order to achieve the same level of performance as an IIR filter. This difference in computational complexity is why IIR filters are often preferred in real-time applications.

An IIR filter loses linear phase due to feedback loops, which can cause phase distortion in the signal. Because FIR filters preserve linear phase, they are useful when signal distortion is unacceptable such as in image processing or data communication.

Some examples of IIR filters:

- **Butterworth filter**: Slow roll-off, maximally flat passband
- **Chebyshev filter**: Faster roll-off, ripples in passband

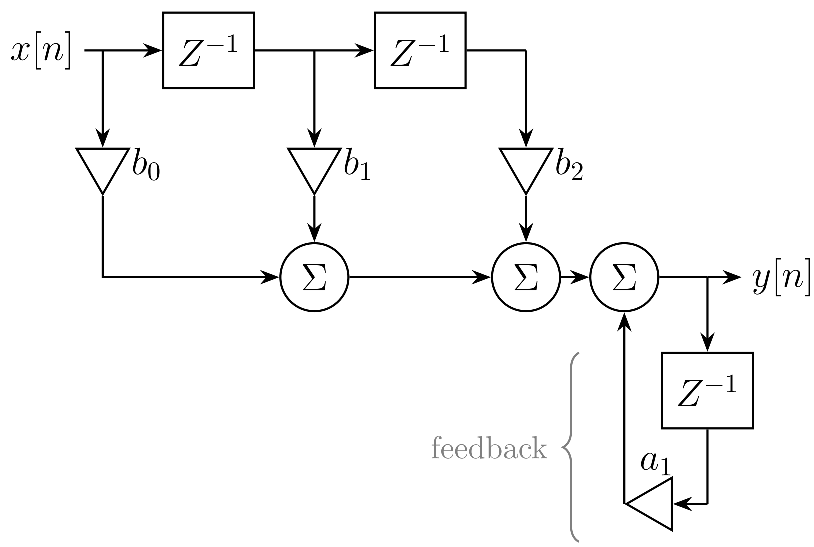
An example of an FIR filter:



Things to note:

- The domain is now discrete (n instead of t) as Z^{-1} expects discrete input.
- This is a 2nd-order FIR filter since there are two Z^{-1} delay blocks.
- The number of **taps** is the number of multipliers b_i, which is always one more than the order of the filter (including for IIR filters).
- The output is the sum of the delayed inputs multiplied by the coefficients.

An example of an IIR filter:



Things to note:

- The domain is also discrete like the FIR filter.
- The diagram is virtually the same as the FIR filter with a feedback component attached.
- It is important to distinguish between the **feedforward** and **feedback** components:
  - The a_i coefficients are the feedback coefficients, and the b_i coefficients are the feedforward coefficients.
  - The Z^{-1} block in the bottom right is a feedback delay block, and the Z^{-1} blocks in the top are feedforward delay blocks.
- This is a 2nd-order IIR filter since there are two feedforward delay blocks and only one feedback delay block, and the maximum of these is two.

13 Random Facts

These random facts may appear on exams to separate an "A" student from the "B" student.

- Turbulence measurements often follow the Rayleigh distribution.
- The mean for the Cauchy distribution is undefined.